Python Installation

Understanding How to Install Python to get started

python™

What is Python Programming?

Python is a high-level, interpreted programming language known for its simplicity, readability, and versatility. It was designed to be easy to understand and write, making it an excellent choice for both beginners and experienced programmers. Python supports multiple programming paradigms, including procedural, object-oriented, and functional programming.

Who developed Python?

Python was developed by **Guido van Rossum** and was first released in 1991. **Van Rossum** developed Python with the intention of creating an easy-to-read language that emphasized code readability and allowed programmers to express concepts in fewer lines of code.

Courtesy: https://gvanrossum.github.io/

Benefits of Python

Readability and Simplicity: Python's syntax is designed to be clean and easy to read, which helps reduce the complexity of writing and maintaining code.

Versatility: Python can be used for a wide range of applications, from web development to data analysis, scientific computing, artificial intelligence, automation, and more.

Large Standard Library: Python comes with a vast standard library that provides tools suited to many tasks, from file I/O to system operations, internet protocols, and more.

Cross-Platform: Python is platform-independent, meaning that it can run on various operating systems, including Windows, macOS, and Linux.

Active Community and Support: Python has a large and active community, offering extensive documentation, tutorials, and third-party packages available through PyPI (Python Package Index).

Integration Capabilities: Python can easily integrate with other languages and technologies, such as C/C++, Java, and .NET, making it useful for a wide range of applications.

No Licensing Fees: Python is free to use, which eliminates the cost barriers associated with proprietary software licenses. This makes it accessible to individuals, startups, educational institutions, and large enterprises alike.

Drawbacks of Python

Slower Execution Speed: Python is generally slower than compiled languages like C or C++ because it is an interpreted language. This can be a disadvantage in performance-critical applications.

High Memory Consumption: Python tends to consume more memory than other programming languages, which can be a concern in environments with limited resources.

Dynamic Typing: While dynamic typing increases flexibility, it can lead to runtime errors that might be caught earlier in statically typed languages.

Not Ideal for Mobile Development: Python is not commonly used for mobile application development, which is more dominated by languages like Java, Kotlin, and Swift.

IDEs of Python

PyCharm: A feature-rich, professional IDE developed by JetBrains. It is widely used for Python development and supports code analysis, debugging, and testing.

VS Code: A lightweight, open-source code editor by Microsoft with robust Python support through extensions.

Jupyter Notebook: An interactive environment that allows you to run Python code in cells and is heavily used in data science and research.

Spyder: An IDE geared towards scientific computing, offering features like an interactive console and data exploration.

IDLE: Python's default IDE that comes with the standard installation of Python. It is simple and suitable for beginners.

Anaconda Navigator: Part of the Anaconda distribution, providing access to various IDEs and tools like Jupyter, Spyder, and RStudio in a single interface.

Thonny: A beginner-friendly IDE that includes a debugger and is simple to use, making it ideal for educational purposes.

Multi-Purpose Programming Language

Python is called a multi-purpose language because it is not limited to a single domain or type of application. It can be used for a variety of programming tasks, including:

Web Development: With frameworks like Django and Flask.

Data Science and Analytics: With libraries like Pandas, NumPy, and Matplotlib.

Machine Learning and AI: With libraries like TensorFlow, PyTorch, and Scikit-learn.

Automation and Scripting: Python is often used for writing scripts to automate repetitive tasks.

Scientific Computing: With libraries like SciPy.

Game Development: With libraries like Pygame.

Network Programming: Python can be used for developing network applications.

Desktop Applications: With frameworks like Tkinter for creating GUI applications.

Ways to Install Python

Using Python website: We can install Python from Python's official website www.python.org however, we normally not prefer to install Python from there as there are many Data Science packages like Numpy, Pandas, Matplotlib, Seaborn, sklearn etc. which we will not get while installing Python from Python website. We have to explicitly install those packages using PIP.

Ways to Install Python

Using Anaconda Distribution: We can install Python from Anaconda's official website that is www.anaconda.com, that is my preferred choice as all the Data Science packages like Numpy, Pandas, Matplotlib, Seaborn, sklearn etc. we will directly get it here. This is one of the most popular ways to install Python in the local machine.

Ways to Install Python

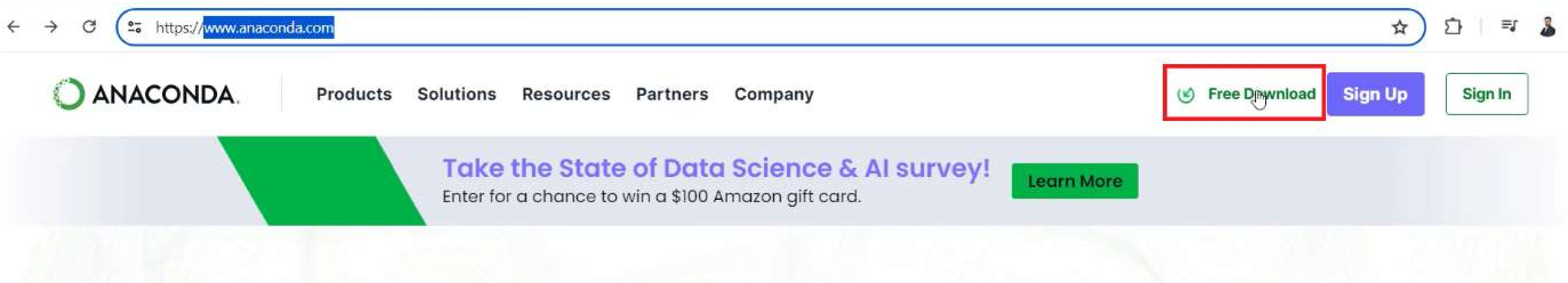
Google Collaboratory: If we don't want to install Python in the local machine and want to use it on the cloud, we can use **https://colab.research.google.com/** which is the online solution given by Google to use Python. This is also one of the most popular ways to play around with Python. The drawback which we have is we have to be dependent on the internet as it is a cloud solution.

What are we going to use?

In this entire sessions we are going to use Anaconda Distribution.

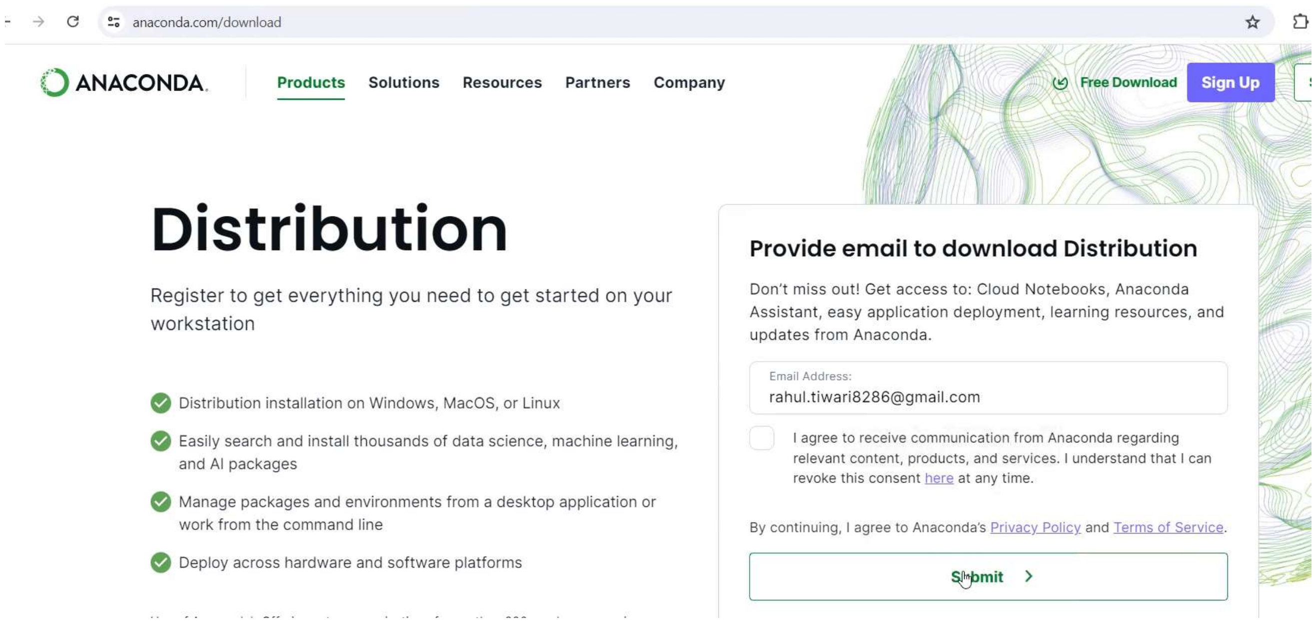
Installation Steps

Step1 : Go to www.anaconda.com and click on free download



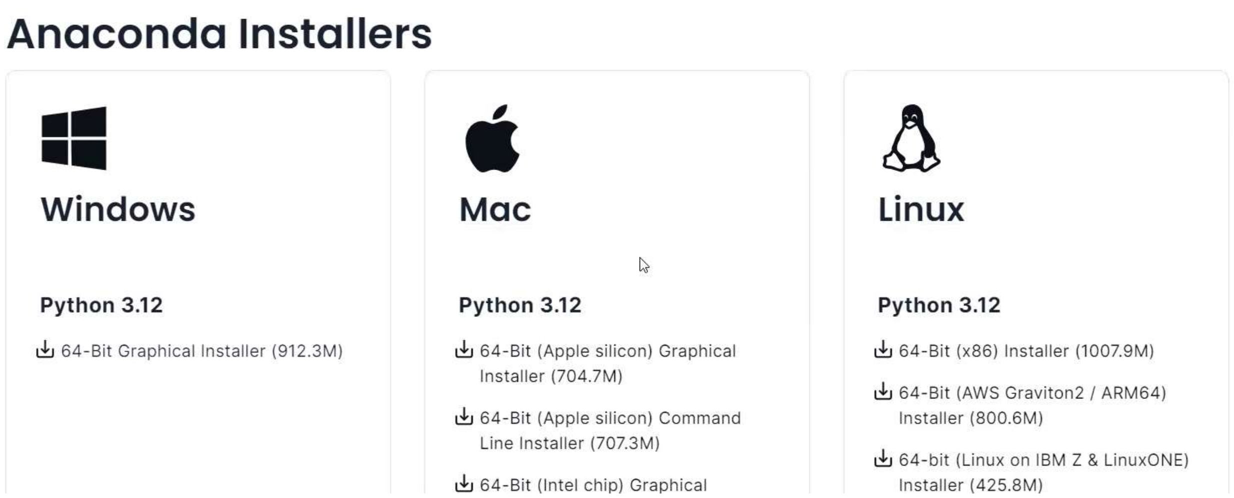
Installation Steps

Step 2: Pass email address and Submit



Installation Steps

Step 3: Install Python based on the operating system, once you click it will install the installer or application file



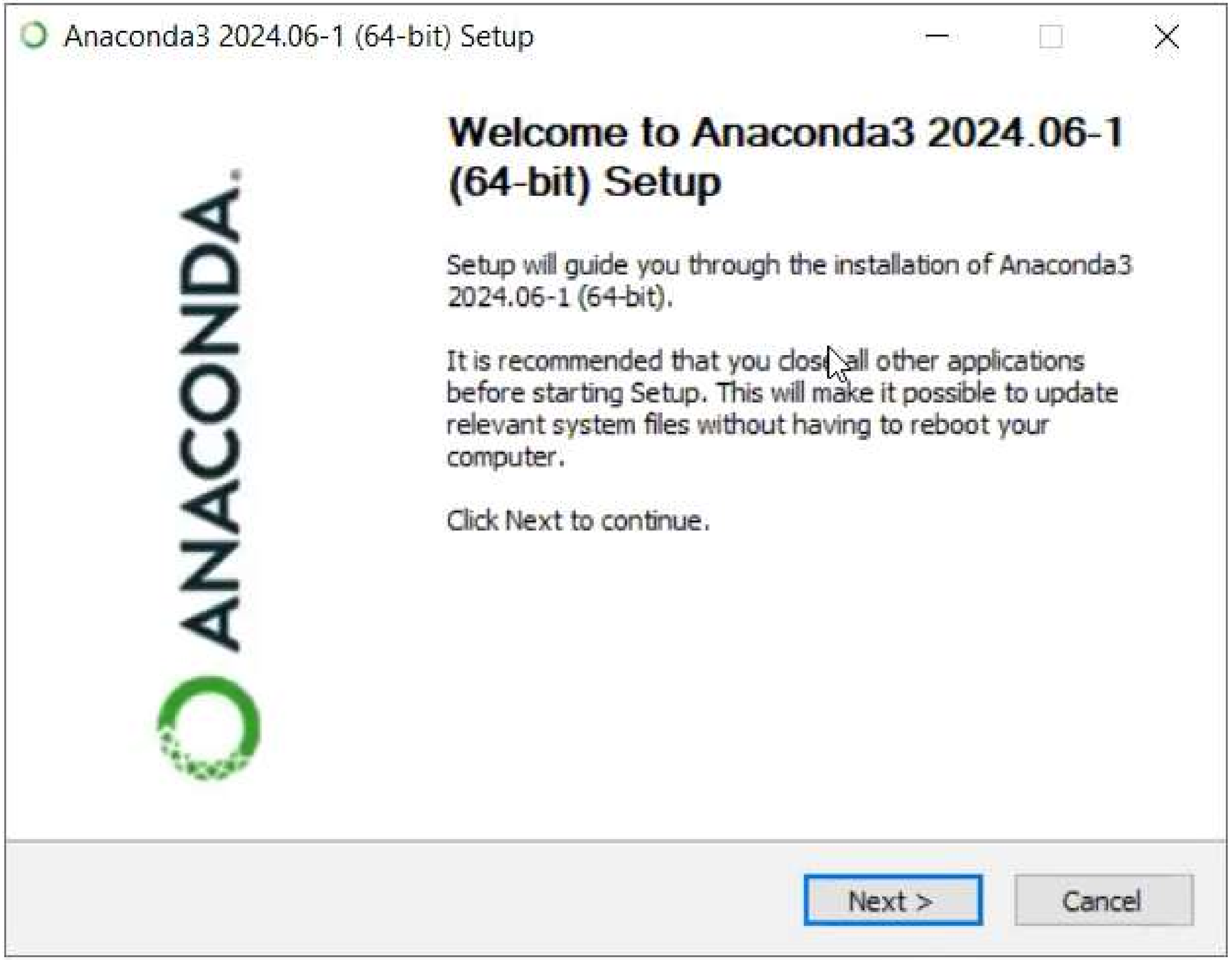
Installation Steps

Step 4: Open this application file and install Python

Anaconda3-2024.06-1-Windows-x86_64.exe	05-07-2024 12:20	Application	9,34,230 KB

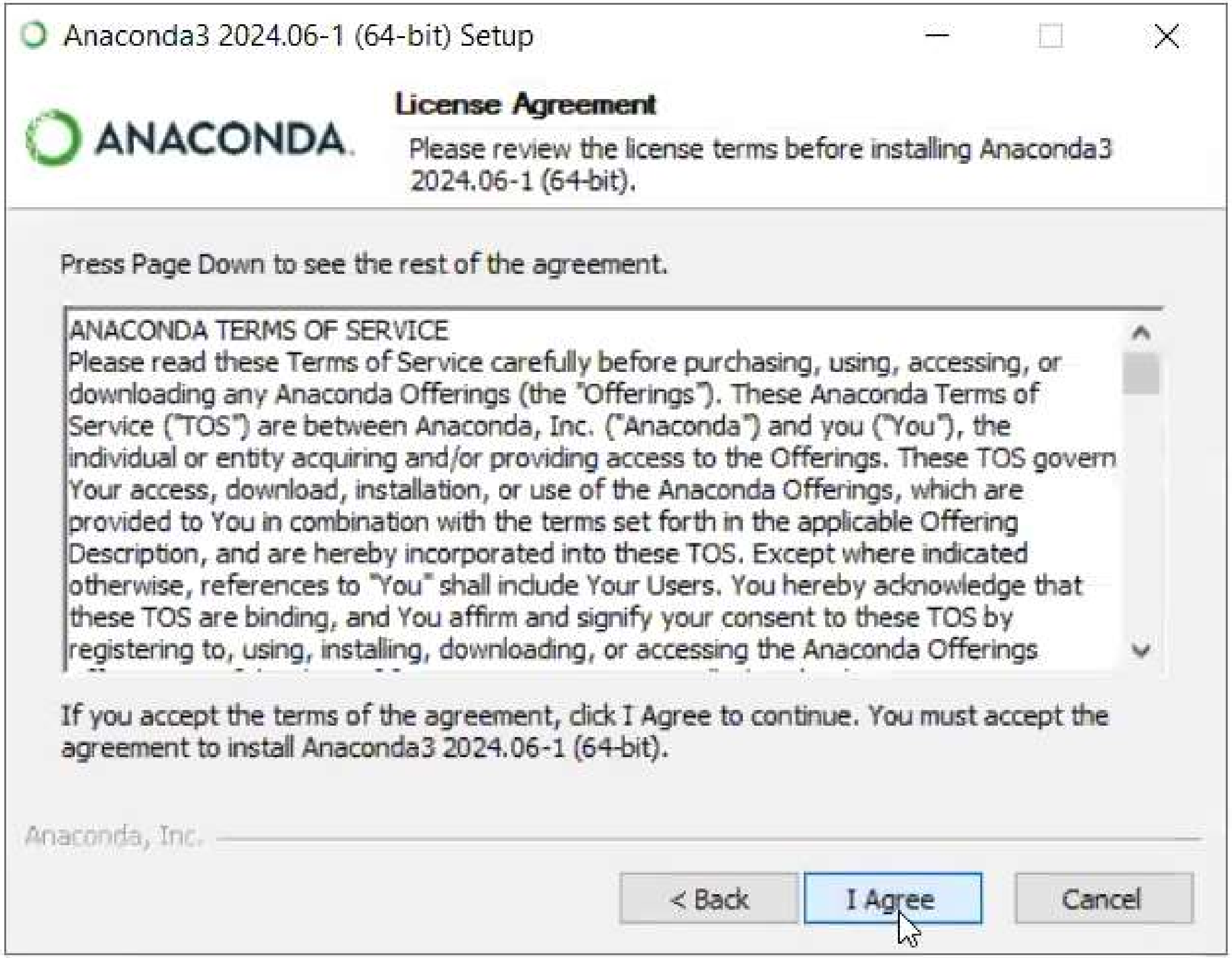
Installation Steps

Step 5: Follow the steps Click Next



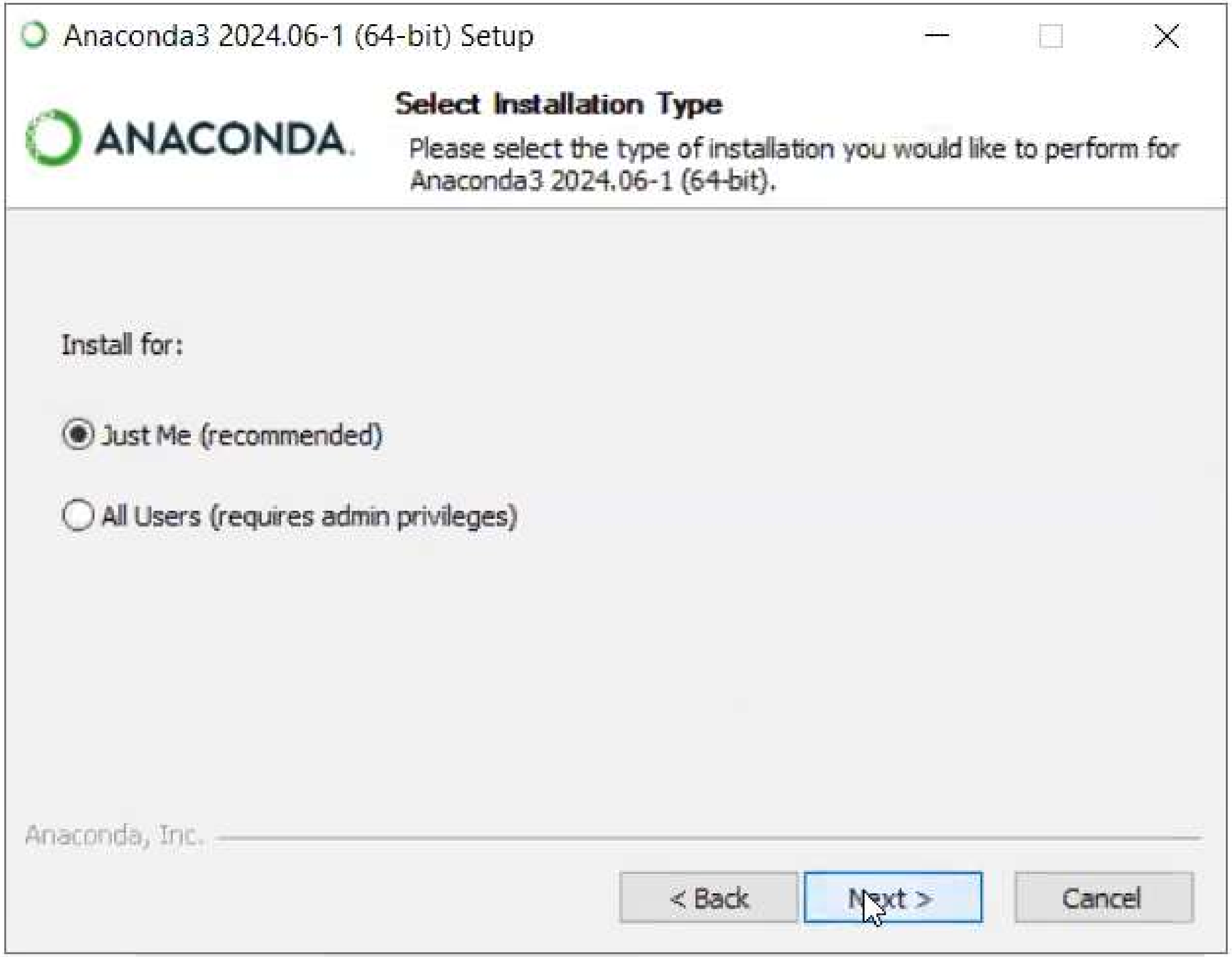
Installation Steps

Step 6: Click I agree



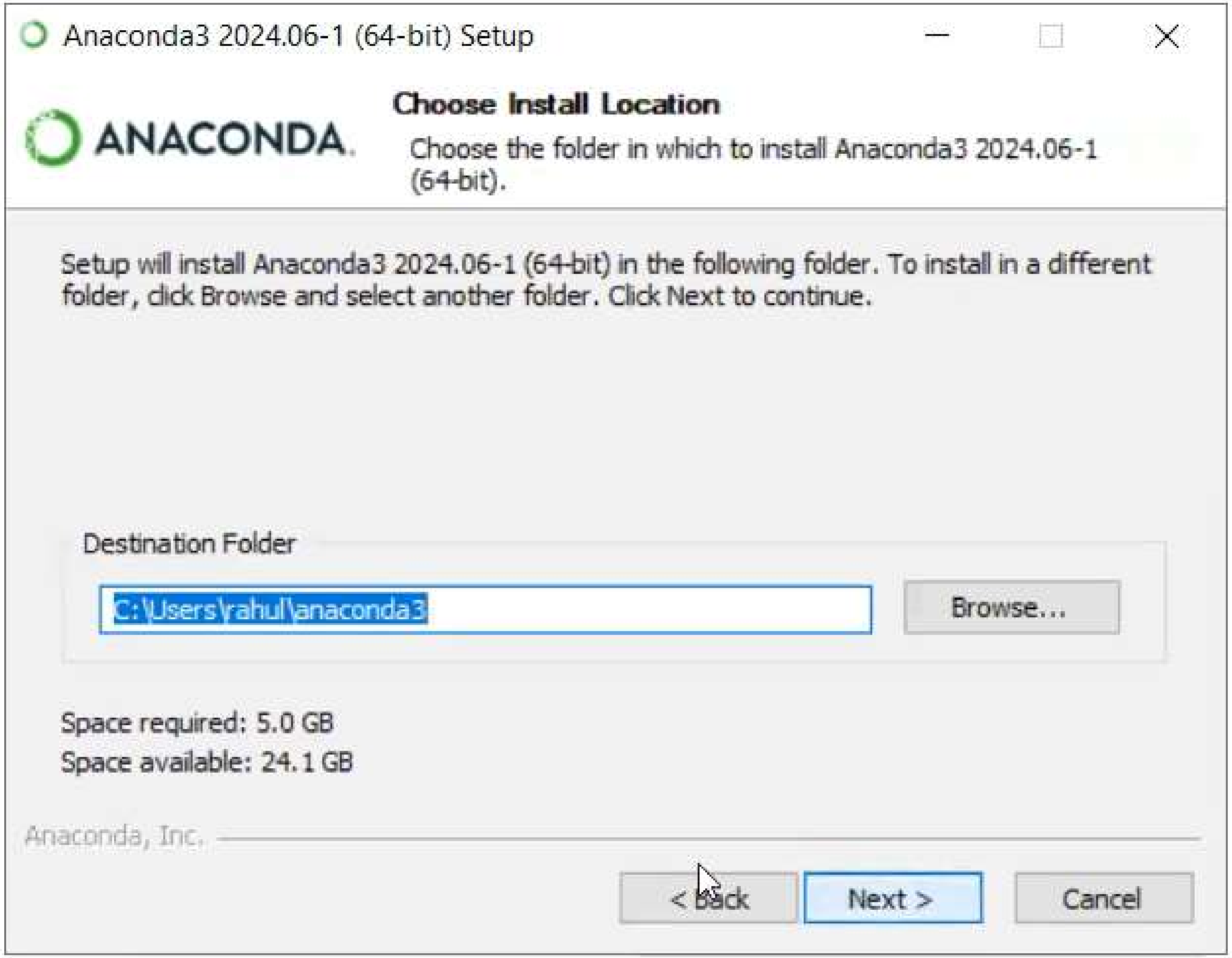
Installation Steps

Step 7: Click Next



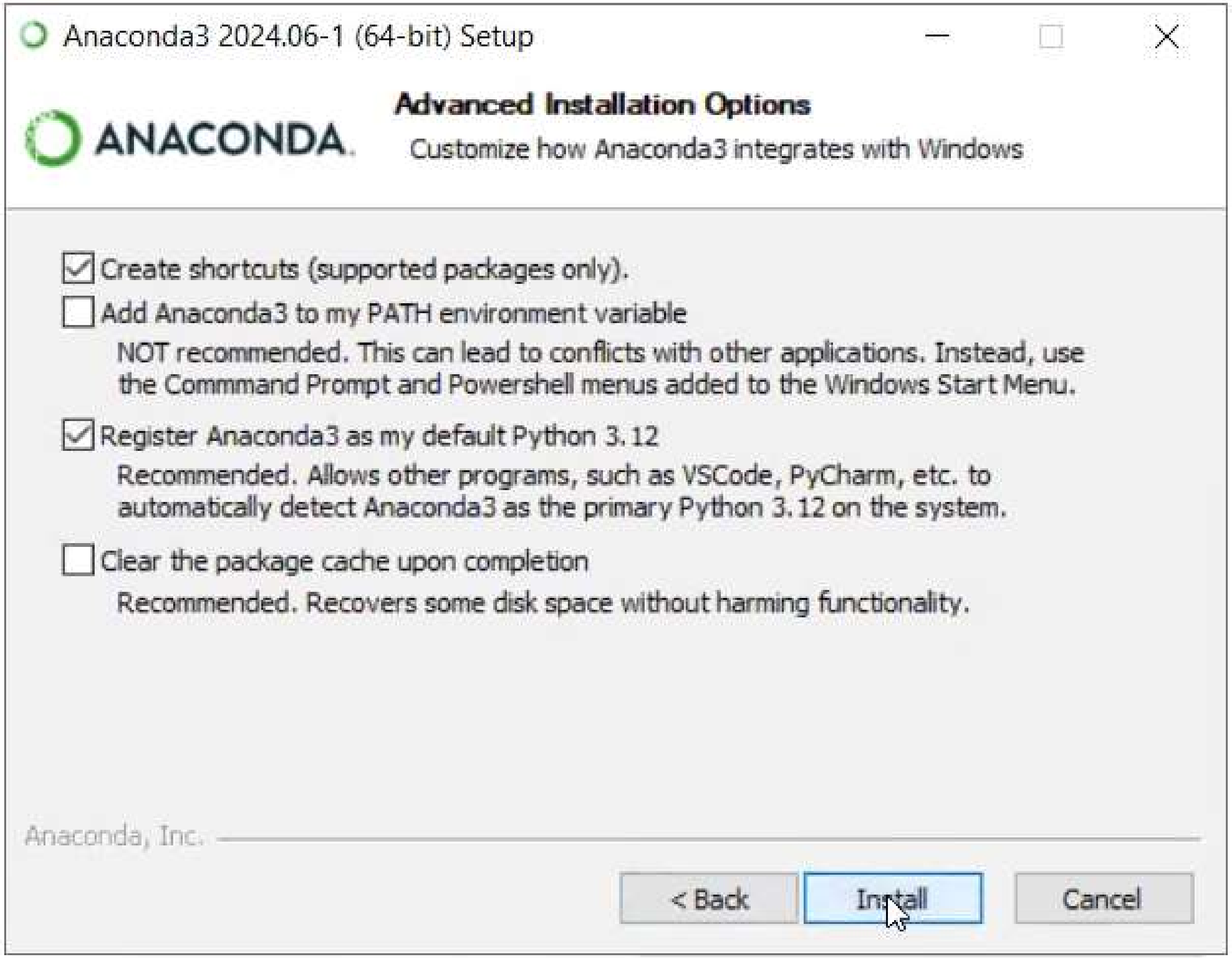
Installation Steps

Step 8: Click Next



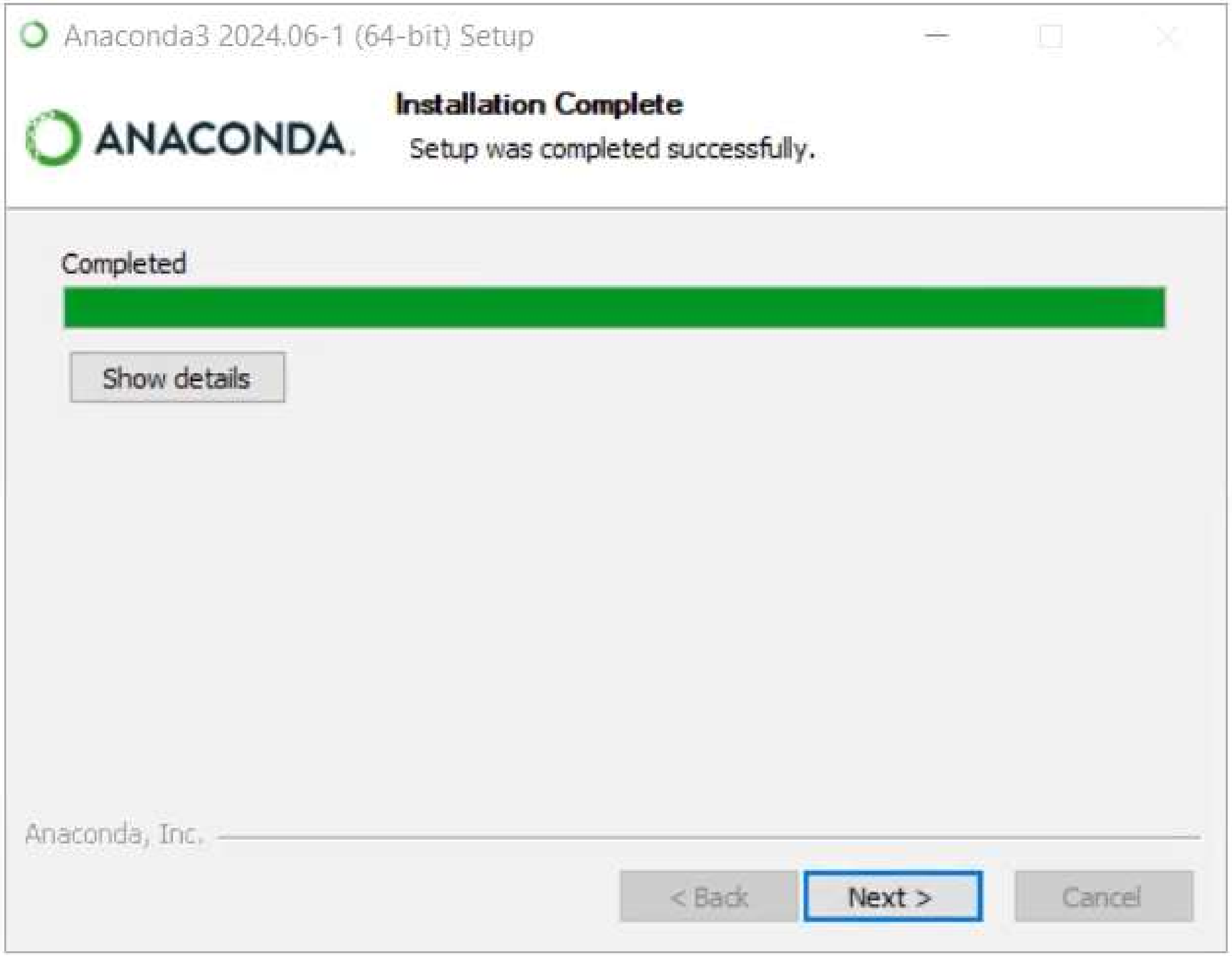
Installation Steps

Step 9: Click Install



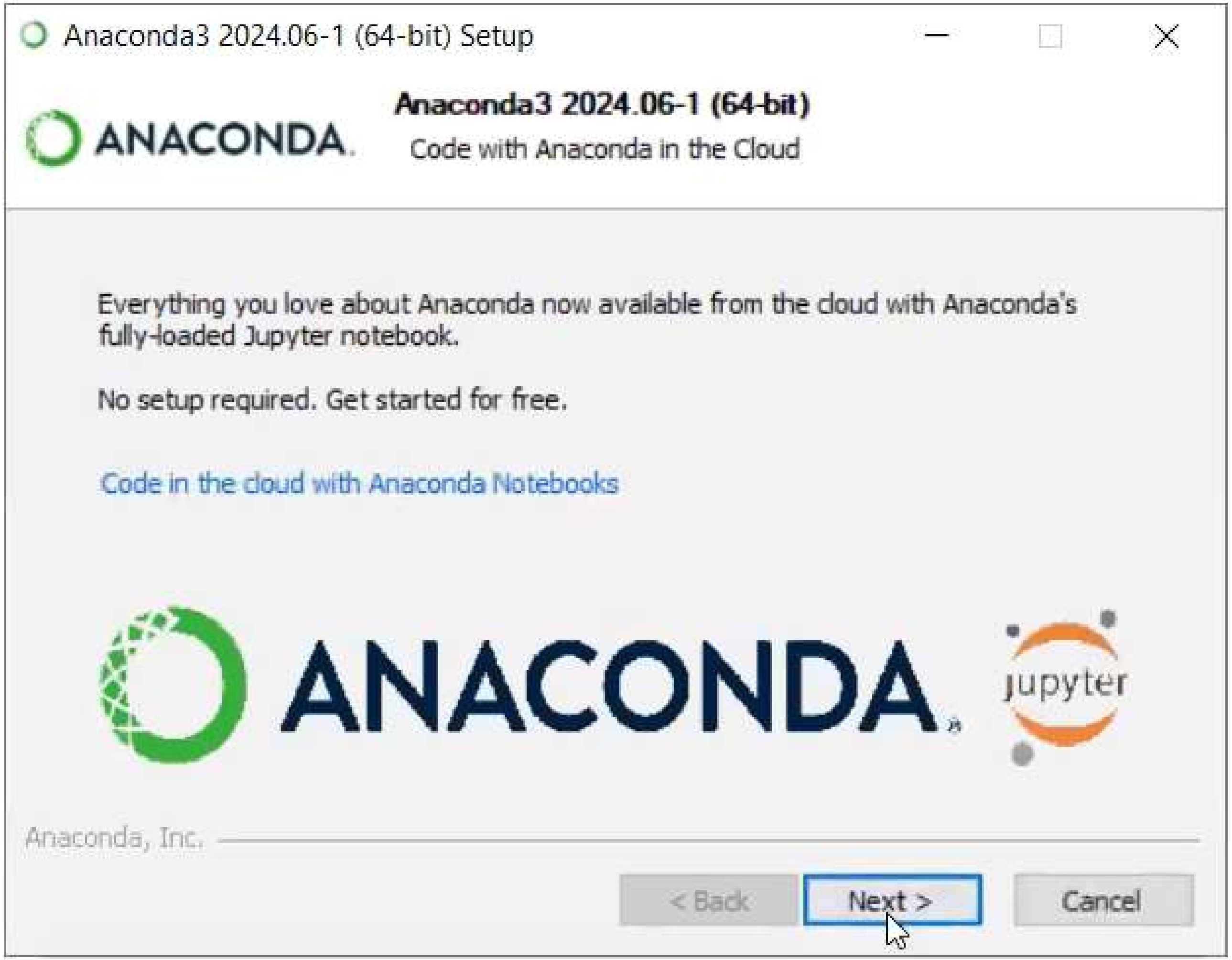
Installation Steps

Step 10: Click Next



Installation Steps

Step 11: Click Next



Installation Steps

Step 12: Click Finish

Practical

How to upload Excel file in Jupyter and Google Colab

What is os Module?

The os module in Python is a standard library module that provides a way to interact with the operating system in a portable and platform-independent manner. It allows you to perform a variety of operating system-related tasks, such as interacting with the file system, manipulating environment variables, and executing system commands.

Key Features and Functions of the os Module

File and Directory Operations: The os module provides functions to perform operations on files and directories, such as creating, removing, and listing directories, and manipulating file paths.

Common Functions:

os.getcwd(): Returns the current working directory.

os.chdir(path): Changes the current working directory to the specified path.

Why we don't need to import os in Google Colab